

As for pre-trained models lacking the ability to follow instructions, Idk-Prompting may not yield satisfactory results.

3.3 Idk Supervised Fine-tuning

Supervised Fine-tuning is a simple yet effective alignment method. We directly use the Idk dataset for Supervised Fine-tuning of the model. Since the Idk dataset contains both questions and responses, this constitutes a conditional generation task. We input the questions into the model and require the model to predict the responses. We perform the standard sequence-to-sequence loss to train our model. SFT details are demonstrated in Appendix B.1.

3.4 Preference-aware Optimization

In this section, we introduce how we conduct preference-aware optimization to help the model perceive its internal knowledge better.

Direct Preference Optimization (DPO) To conduct DPO, we first train a SFT model on half of the Idk dataset as a warm up, then we collect responses from this SFT model on the other half of the Idk data. For a given question, we conduct random sampling to gather multiple responses. Finally, we construct preference data based on these generated responses. We demonstrate the procedure in Figure 3 (bottom). A preference data sample consists of a question, a chosen response, and a rejected response. The questions in the Idk dataset can be categorized into two types: those the model knows and those it does not know. For questions the model knows, we use the correct response generated by it as the chosen response and “I don’t know” as the rejected response. For questions the model does not know, we use “I don’t know” as the chosen response and its incorrectly generated response as the rejected response. Besides, we find that only using the DPO loss Rafailov et al. (2023) can occasionally result in the model’s inability to accurately generate the Idk template. Therefore, in addition to the original DPO loss, we also incorporate SFT loss for the chosen responses and multiply it by a coefficient α . The details of the DPO are demonstrated in Appendix B.2.

Best-of-n Sampling (BoN) We also try to determine if the model knows the answer to a certain question by training a reward model to score the candidate responses. We first train a SFT model using a half of the Idk data and then use the SFT model to initialize the reward model. After collecting responses on the other half of the Idk dataset and constructing preference data using the same procedure as 3, we train the reward model using a pairwise loss. During inference, we employ the Best-of-10 strategy. First, we sample ten responses using the SFT model, then we score these candidate responses with the reward model. The response with the highest reward score is selected as the final response. The details of reward modeling are demonstrated in Appendix B.3.

Proximal Policy Optimization (PPO) Based on our reward model, we can use proximal policy optimization to optimize the model. We use the same inputs for PPO training as we do for reward modeling, but sample responses in an online manner. The details of the PPO are demonstrated in Appendix B.4.

Hindsight Instruction Relabeling (HIR) So far, our Idk dataset is constructed based on a fixed Ik threshold. In order to utilize all Idk datasets constructed with different Ik thresholds, inspired by Hindsight Instruction Relabeling (Zhang et al., 2023a), we design an instruction format to relabel all Idk datasets. Specifically, we prepend the following instruction to each question in the Idk datasets:

Your current knowledge expression confidence level is <X>, please answer the user’s question: <Question>

where < Question > is a question from an Idk dataset and < X > is the value of model’s knowledge confidence level ranging from 0 to 1.0, derived from the Ik threshold corre-

sponding to the Idk dataset. The lower the knowledge confidence level, the more inclined the model is to refuse answering questions. Then we use the combined Idk dataset to perform supervised fine-tuning. The advantage of using instruction relabeling is that we can control the model to adopt either a conservative or aggressive response strategy through the instruction, without the need to retrain the model. The details of HIR are demonstrated in Appendix B.5.

4 Experiments

4.1 Dataset

TriviaQA (Joshi et al., 2017) is a reading comprehension dataset, but its question-answer pairs can be used for open-domain question answering tasks. We use the training set of TriviaQA, consisting of 87,622 samples, to construct the training and development sets of the Idk dataset. Since TriviaQA does not provide ground truth for the test set, we use the development set of TriviaQA to construct the test set for the Idk dataset, which comprises a total of 11,313 samples. The detailed statistical information of the Idk datasets is provided in Appendix A.

Additionally, we use the Natural Questions (NQ) (Kwiatkowski et al., 2019) and ALCUNA (Yin et al., 2023a) datasets as the out-of-distribution (OOD) questions. NQ is a question answering dataset consisting of real queries issued to the Google search engine. We use the development set of NQ-Open, which contains 3,610 samples, to construct the OOD test set. According to the experimental results in Wang et al. (2023a), using lexical matching for the automatic evaluation on the development set of the NQ dataset shows more than 80% consistency with human expert assessments. Therefore, we use lexical matching to judge the correctness of model answers and to label “I don’t know” responses.

ALCUNA is a benchmark to assess LLMs’ abilities in new knowledge understanding. It creates new artificial entities by altering existing entity attributes and generates questions about these artificial entities. Since these entities are artificially created, the model cannot possibly possess this knowledge. Therefore, we use a portion of the questions from ALCUNA to test whether the model can refuse to answer, totaling 8,857 samples.

4.2 Metrics and Evaluation

We report the following metrics.

- **IK-IK Rate:** I know what I know (Ik-Ik) rate represents the proportion of questions answered correctly by the model out of all questions.
- **IK-IDK Rate:** I know what I don’t know (Ik-Idk) rate represents the proportion of questions that the model correctly refuses to answer, out of all questions.
- **TRUTHFUL Rate:** Truthful rate is the sum of Ik-Ik rate and Ik-Idk rate. It represents the proportion of questions for which the model provides truthful responses. The higher the value of TRUTHFUL rate, the clearer the model’s perception of what it knows and does not know, which also indicates a higher level of truthfulness. The TRUTHFUL value of an ideal truthful model should ideally reach 100%.

The higher these three metrics are, the better. Among these metrics, we argue that the TRUTHFUL rate is the most important one because it indicates the probability that users will receive a truthful response.

To calculate these metrics, we categorize the inference results into four knowledge quadrants using the following method.

- **IK-IK:** If a question model does not refuse to answer and the answer is correct, then the question belongs to the Ik-Ik category. We determine whether the model’s answer is correct by checking if the ground truth appears in the model’s response.

- IK-IDK: If a question model refuses to answer, and the question is marked as one that the model does not know, then this question belongs to Ik-Idk category. We determine whether the model refuses to answer a question by checking whether the refusal template appears in the model’s response.
- IDK-IK: If a question model refuses to answer, but the question is not marked as one the model does not know, then this question falls into the Idk-Ik category.
- IDK-IDK: If a question model does not refuse to answer but provides an incorrect response, then the question belongs to the Idk-Idk category.

We use Llama-2-7b-chat as our initial model for further training, with specific training details introduced in Appendix B. We test the trained model on the test set of the Idk dataset to evaluate whether the model can distinguish between questions it knows and does not know. Except for Idk-BoN, we use greedy decoding in all tests. For Idk-BoN, we set the temperature coefficient to 1.0 and top_p to 0.9, sample ten responses, and then score them using the reward model. The response with the highest reward score is selected as the final model response.

4.3 Main Results

	TriviaQA			Natural Questions			ALCUNA
	IK-IK	IK-IDK	TRUTHFUL	IK-IK	IK-IDK	TRUTHFUL	IK-IDK
Idk-Dataset _{test}	45.05	54.95	100.00	24.65	75.35	100.00	100.00
Idk-Prompting	37.36	29.58	66.93	19.75	41.72	61.47	91.67
Idk-SFT	28.57	46.19	74.75 ^{↑7.82}	15.93	53.99	69.92 ^{↑8.45}	98.01
Idk-DPO	39.30	38.59	77.89 ^{↑10.96}	20.91	45.60	66.51 ^{↑5.04}	98.08
Idk-BoN _{N=10}	38.37	40.59	78.96 ^{↑12.03}	20.55	47.40	67.95 ^{↑6.48}	98.32
Idk-PPO	35.90	40.57	76.47 ^{↑9.54}	23.13	42.08	65.21 ^{↑3.47}	92.66
Idk-HIR	27.36	48.55	75.91 ^{↑8.98}	15.40	56.90	72.30 ^{↑10.83}	98.96

Table 1: Overall results on the test set of the Idk dataset constructed based on TriviaQA and out-of-distribution test sets.

The overall results are in Table 1. The Idk-Dataset used for evaluation contains 45.05% IK-IK questions and 54.95% IK-IDK questions, which can be seen as two upper bounds of IK-IK and IK-IDK rate. Simply using an Idk prompt to let the model refuse to answer questions it doesn’t know can have a certain effect, but the model’s TRUTHFUL rate is still only 66.93%. The Idk-SFT can increase the TRUTHFUL rate to 74.75%, but this will result in a decrease in the IK-IK rate, which can be considered a form of “alignment tax”. We find that preference optimization can encourage the model to answer questions, thereby mitigating the alignment tax. DPO³, PPO, and BoN can all reduce the loss of IK-IK while maintaining a relatively high IK-IDK rate. Idk-BoN achieves the highest TRUTHFUL rate. Idk-HIR combines all Idk datasets, which can improve IK-IDK rate but help less for IK-IK rate. However, Idk-HIR provides an switching method for Ik-threshold that does not need to retrain the model. Overall, by aligning with the Idk dataset, we can transform IDK-IK and IDK-IDK questions into IK-IK and IK-IDK questions. The model can have a clear perception of whether it knows the answers to most questions in the test set, significantly increasing truthfulness compared to before the alignment. The additional experimental results are represented in Appendix C.

Evaluation on out-of-distribution data We also test whether the aligned model is capable of refusing to answer questions it does not know when encountering out-of-distribution (OOD) data. We first construct the Idk dataset for testing based on Natural Questions, setting the Ik threshold to 1.0. As shown in Tabel 1, the Idk dataset contains 24.65% IK-IK

³We find that the DPO model, when refusing to answer questions within ALCUNA, occasionally rephrases our Idk template. Consequently, we utilize a substring of the original Idk template: “I am not sure what the answer is” to detect whether the model refuse to answer the question.